ModelIt!

Student Activity Pack

Why Do Athletes Collapse in the Heat?

Modeling Homeostasis and the Body's Breaking Point

Grade 09L1 | NGSS HS-LS1-3 | Stability and Change

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

Multiple Choice: Circle the best answer for each question.

1 A student claims that sweating is a sign of being out of shape. Which of the following best explains why this claim is scientifically inaccurate?

- **A.** Sweating is the body's primary mechanism for evaporative cooling and is more efficient in well-conditioned athletes
- **B.** Sweating only occurs when the body is overheating due to illness
- **C.** Sweating is controlled by voluntary muscles and can be stopped at will
- **D.** Sweating depletes the body's salt reserves, which is harmful regardless of fitness level

2 During a football game on a hot day, two players of similar fitness levels are performing at the same intensity. Player A collapses from heat exhaustion while Player B continues playing. Which factor most likely explains this difference?

- **A.** Player A has a genetic condition that prevents all sweating
- **B.** Player A consumed significantly less water before and during the game than Player B
- **C.** Player B is naturally immune to heat-related illness
- **D.** Player A was wearing a darker uniform color that absorbed more heat

3 Which of the following best describes the concept of homeostasis?

- **A.** The body's ability to increase temperature during exercise to improve performance
- **B.** A state where the body's internal conditions remain perfectly constant at all times
- **C.** The body's ability to maintain stable internal conditions within a functional range despite changing external conditions
- **D.** A process that only occurs when the body is at rest and not under physical stress

4 An athlete's core body temperature rises from 37°C to 39°C during a workout. The body responds by increasing sweat production, which eventually brings the temperature back toward 37°C. This process is an example of which type of biological mechanism?

- **A.** Positive feedback, because the response amplifies the original change
- **B.** Negative feedback, because the response counteracts the original change
- **C.** Passive diffusion, because heat naturally moves from hot to cold
- **D.** Active transport, because the body uses energy to move heat molecules

5 **A coach notices that heat-related illness incidents at their school have occurred at outdoor temperatures as low as 78°F (26°C). Which explanation best accounts for this observation?**

- **A.** Heat illness can only occur above 100°F, so these incidents must have been misdiagnosed
- **B.** Air temperature alone does not determine heat illness risk; humidity, exertion level, and hydration status interact to determine vulnerability
- **C.** The thermometers used to measure temperature were likely inaccurate
- **D.** Heat illness at low temperatures is caused by an allergic reaction to sunlight rather than thermoregulatory failure

1 **Why do you think some athletes collapse from the heat while others on the same team are fine?**

2 **What happens inside your body when you get too hot during exercise?**

3 **Draw a diagram showing how you think the body cools itself during intense physical activity.**

4 **What is homeostasis and why is it important for survival?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Core Temperature | Sweat Rate | Hydration Level | Physical Exertion

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Moderate & Hydrated

Moderate exertion, high hydration

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Intense & Dehydrated

Maximum exertion, low hydration

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Gradual Dehydration

Steady exertion, declining hydration

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Ambient Temperature — The external air temperature and humidity, which affects how efficiently sweat can evaporate and cool the body

Blood Volume — The total volume of blood circulating in the body, which decreases with dehydration and reduces the body's ability to transport heat

Acclimatization Level — How adapted an athlete's body is to exercising in heat, which improves sweating efficiency and reduces strain

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why do some athletes collapse during games while others playing the same sport are fine? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate that homeostasis depends on negative feedback loops?**

2. **What happens in your model when the feedback loop is broken by dehydration?**

3. **Why might two athletes of the same fitness level respond differently to the same heat conditions?**

4. **How could coaches use your model to design safer practice schedules?**

5. **What are the limitations of modeling a complex biological system with only four components?**

8 STEM Challenge: Design a Heat Safety Alert System

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YOUR ENGINEERING MISSION

Design a wearable or sideline monitoring system that tracks athlete biometrics and alerts coaches before heat-related illness occurs.

THE SCENARIO

A high school athletic department has had three heat-related incidents in two years. The principal has hired your team to design a monitoring system that can predict when an athlete is approaching dangerous territory and alert the coaching staff before collapse occurs.

Engineering Design Thinking

1. WHAT SENSORS WOULD YOU NEED AND WHERE WOULD THEY BE PLACED ON THE ATHLETE?
2. HOW WOULD THE SYSTEM COMMUNICATE ALERTS TO COACHES DURING A GAME?
3. HOW WOULD YOU ACCOUNT FOR DIFFERENCES IN BODY SIZE, FITNESS LEVEL, AND ACCLIMATIZATION?
4. WHAT DATA WOULD YOU DISPLAY ON THE COACH'S DASHBOARD?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Sports Physiologists study how the body responds to physical activity and environmental stress. They work with professional sports teams, military programs, and research institutions, earning $60,000–

$120,000/year.

9 Post-Assessment

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Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 In a computational model of thermoregulation, a student sets Physical Exertion to maximum and Hydration Level to low. The model shows Core Temperature rising sharply while Sweat Rate initially increases then drops to near zero. Which explanation best interprets this model behavior?

- **A.** The model is malfunctioning because sweat rate should continue to increase with temperature
- **B.** The negative feedback loop has broken down because the body cannot produce sweat without adequate water, causing an uncontrolled temperature increase
- **C.** The body has switched from negative feedback to a more efficient cooling mechanism
- **D.** Sweat rate decreases because the body has successfully cooled itself and no longer needs evaporative cooling

2 A student's model shows that during the 'Gradual Dehydration' scenario, core temperature remains stable for the first 45 minutes, then begins rising rapidly. Which concept best explains this pattern?

- **A.** The system has a tipping point where the feedback loop transitions from maintaining homeostasis to failing catastrophically
- **B.** The body's thermostat resets to a higher temperature after 45 minutes of exercise
- **C.** The model has a built-in delay that prevents temperature changes for the first 45 minutes
- **D.** Core temperature always rises linearly during exercise regardless of hydration

3 Two students build models of thermoregulation. Student A's model includes only Core Temperature and Sweat Rate. Student B's model adds Hydration Level and Physical Exertion. Which statement best evaluates the difference between these models?

- **A.** Student A's model is superior because simpler models are always more accurate
- **B.** Student B's model can predict system failure conditions that Student A's model cannot, because it includes the external inputs that drive and constrain the feedback loop
- **C.** Both models are equally valid because they contain the same core feedback mechanism
- **D.** Neither model is useful because thermoregulation requires at least ten components to model accurately

4 Based on model evidence, which intervention would be most effective at preventing heat-related illness during a high-intensity outdoor practice in 90°F weather?

- **A.** Moving practice to a shaded area, because shade eliminates the risk of overheating
- **B.** Implementing mandatory hydration breaks every 15 minutes, because maintaining hydration preserves the integrity of the negative feedback loop
- **C.** Reducing practice duration to 30 minutes, because heat illness cannot occur in short time periods
- **D.** Having athletes wear lighter-colored clothing, because clothing color is the primary determinant of core temperature

5 **A student extends their thermoregulation model by adding an Acclimatization Level component. They find that higher acclimatization shifts the tipping point, allowing athletes to exercise longer before the feedback loop fails. Which reasoning best explains this model prediction?**

A. Acclimatized athletes do not produce body heat during exercise

B. Acclimatization improves sweating efficiency and plasma volume, allowing the cooling system to operate effectively at higher core temperatures and lower hydration levels

C. Acclimatization eliminates the body's need for water during exercise

D. Acclimatized athletes have a lower resting core temperature, so they can tolerate unlimited heat exposure